HDOC - 59TH DAY

LONGEST UNIQUE SUBSTRING

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Problem Statement

Write a C program that reads a string s and passes its address to a function named longest_unique_substring() using pointers. The function must:

- Find the longest substring containing no repeated characters.
- Calculate and display the length of the longest unique substring.
- Treat uppercase and lowercase letters as different characters.
- Display 0 if the input string is empty.

Algorithm

1. Start the program.
2. Declare a character array s in main().
3. Read the string using fgets() and remove the newline character.
4. Pass the address of s to longest_unique_substring().
5. Initialize max = 0 and declare pointer variables p, q and r.
6. If the string is empty, display 0 and return.
7. Use p as the starting position of a substring and q to move through its characters.

8. Use r to check whether the current character has already appeared in the substring.

9. If a repeated character is found, stop checking the current substring.

10. Otherwise, increase len and update max whenever len is greater than max.

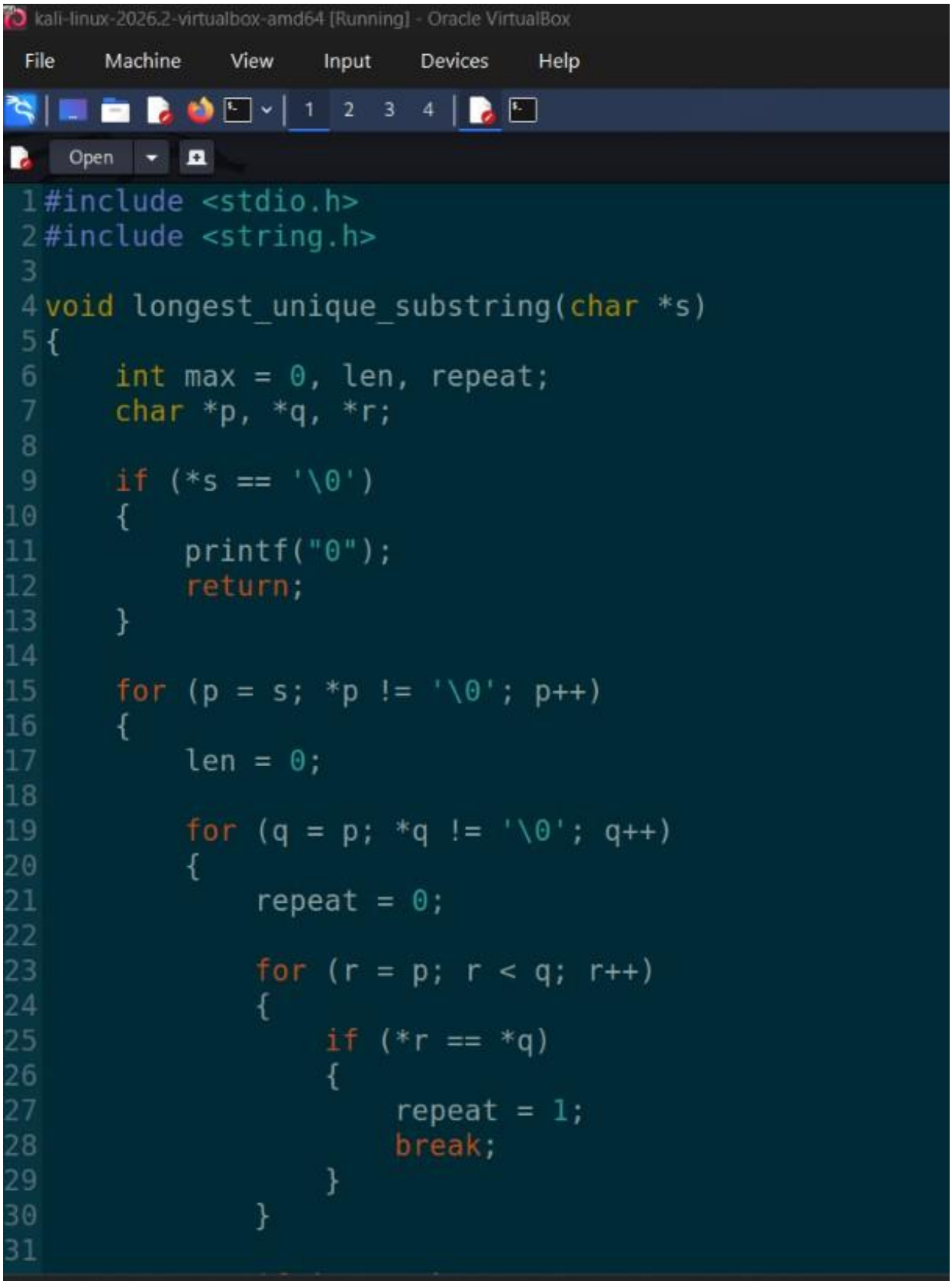
11. Display max as the length of the longest unique substring.

12. Stop.

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	abcabcbb	3	3	Pass
2	pwwkew	3	3	Pass
3	abcdef	6	6	Pass
4	aA	2	2	Pass
5	Empty string	0	0	Pass

C Program




```c
#include <stdio.h>
#include <string.h>

void longest_unique_substring(char *s)
{
    int max = 0, len, repeat;
    char *p, *q, *r;

    if (*s == '\0')
    {
        printf("0");
        return;
    }

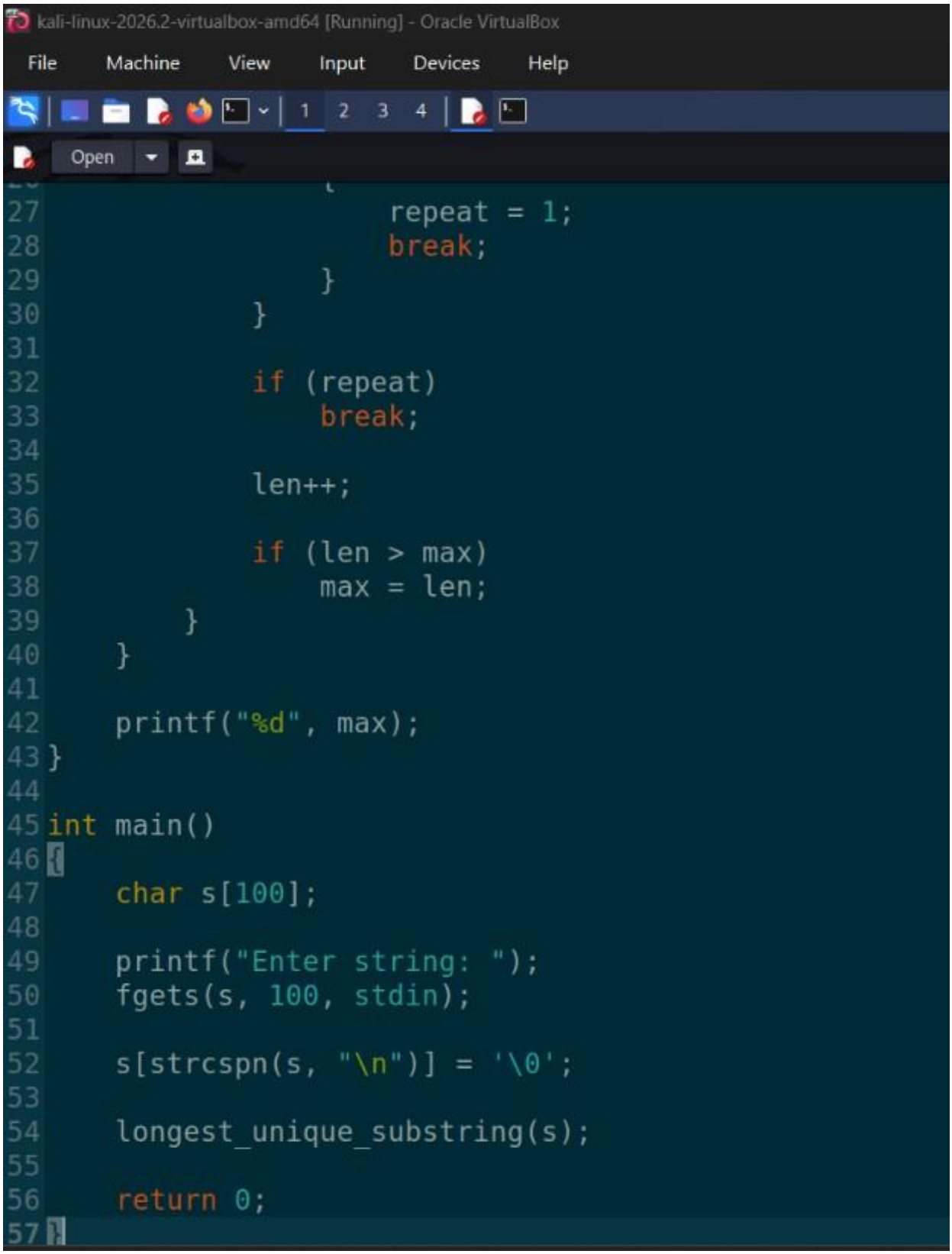
    for (p = s; *p != '\0'; p++)
    {
        len = 0;

        for (q = p; *q != '\0'; q++)
        {
            repeat = 0;

            for (r = p; r < q; r++)
            {
                if (*r == *q)
                {
                    repeat = 1;
                    break;
                }
            }

```

C Program (Continued)




```
                repeat = 1;
                break;
            }
        }

        if (repeat)
            break;

        len++;

        if (len > max)
            max = len;
    }
}

printf("%d", max);
}

int main()
{
    char s[100];

    printf("Enter string: ");
    fgets(s, 100, stdin);

    s[strcspn(s, "\n")] = '\0';

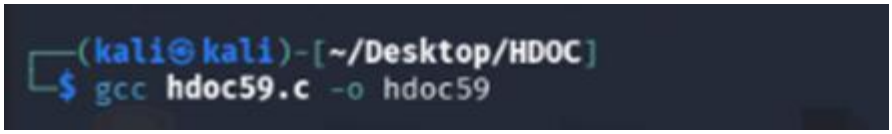
    longest_unique_substring(s);

    return 0;
}
```

Compilation and Execution

Compile the program using:

gcc hdoc59.c -o hdoc59



Execute the program using:

./hdoc59

